Price Multipliers are Larger at More Aggregate Levels

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Increased interest in price impact at low frequencies

- Increased interest in **quantity questions** in asset pricing:
  - Holdings constant fundamental value of a security/portfolio, what is the **quantitative** impact of buying 1% of outstanding supply?
- The typical regression: $\Delta P = M * \Delta Q + \epsilon$
  - Great variation in multiplier: $M \approx 0.5 - 8$ in existing literature using various demand shocks ΔQ.[2]
- Typical interpretation/hypothesis: Larger M suggests fewer available substitutes.
  - Smallest multiplier for individual stocks (many substitutes), larger for aggregate factors (fewer substitutes), largest for aggregate stock market (no substitutes?)
  - Has not been explicitly tested yet in one coherent framework!

[2] Index Inclusions/Reconstitutions (Pavlova & Sikorskaya, 2022) Dividend Reinvestments (Salomon & Hartzmark, 2022) Mutual fund fire sales (Coval & Stafford, 2008) Flow-driven trades (Lou 2012) GIV (Gabaix & Koijen, 2023) Ratings-driven Demand (Ben-David et al, 2024)

This paper

1. Propose "new" demand shock: Signed order flow summed over time.
2. Study multiplier at different levels of aggregation using same demand shock & econometric specification

- Decompose demand ΔQ into idiosyncratic and group-level component $\Delta Q_{idio} + \Delta Q_{group}$ and run decomposed regression

$$\Delta P = M_{idio} * \Delta Q_{idio} + M_{group} * \Delta Q_{group} + \epsilon$$

- Price multipliers larger at more aggregated levels: $M_{group} > M_{idio}$
- Intuitively: Buying 1% of all value stocks has higher impact (in %) than buying 1% of an individual value stock.

Comments

- **Great!**
  1. New demand measure brings **market microstructure** and **macro asset pricing** together. Exciting new links!
  2. Very robust empirical finding. Holds for alternative demand shocks, different groupings, frequencies etc.
- Small Comments
  1. Is signed order flow nonfundamental/uninformed?
  2. Link to Chaudhary, Fu & Li, 2023:
     "Corporate Bond Multipliers: Substitutes Matter"
  3. Interpretation of the findings: Arbitrage Risk versus Constraints
  4. Calibration of CARA model: Volatility of Cashflows vs Returns

1. Is signed order flow nonfundamental/uninformed?

- At the individual trade level (milliseconds), quantity of new information released probably small.
  - In trade-level regression potential bias from information negligible
  - When aggregating to lower frequencies not so trivial any more
- Paper uses *total* signed order flow (as opposed to order flow from index-rebalancing, dividend reinvestment etc). Is all aggressive trading uninformed? Probably not.
  - Perhaps phrasing "informed vs uninformed" is not ideal. Rather investor-specific demand shifts and common demand shifts.
  - Investor-specific demand shifts have greater impact at more aggregate levels. Striking!
- More generally: Is positive multiplier from order flow mechanical? What moves prices **other than** order flow?
  - At low frequency: Common demand shifts/agreement $\rightarrow$ no trading
  - At high frequency (tick-level): Quote adjustments.[3]
  - Positive multiplier = Price discovery?

[3] Btw: CRSP prices are not mid-points between quotes, but executed orders, so in some sense the only thing that can move CRSPs prices is aggressive trades.

2. Link to Chaudhary, Fu & Li, 2023: "Corporate Bond Multipliers: Substitutes Matter"

- Both papers find higher multipliers at higher levels of aggregation.
  1. This paper: Equities

  $$\Delta P_{it} = M_{idio} * \Delta Q_{it} + M_{group} * \Delta Q_{group(i),t} + \epsilon_{i,t}$$

  2. Chaudhary, Fu & Li (2023): Corporate Bonds

  $$\Delta P_{it} = M_{idio} * \Delta Q_{it} + M_{group} * \Delta P_{group(i),t} + \epsilon_{i,t}$$

  ... using $\Delta Q_{group(i),t}$ as an instrument for $\Delta P_{group(i),t}$.
- I think the papers are complementary, but a reference (at least in a footnote) would be useful.
  - In one paper M_{group} identifies cross-multipliers, in the other it identifies (scaled) cross-elasticities.
  - In CARA model M_{group} captures the off-diagonal elements of versus $\frac{1}{\gamma}\Sigma^{-1}$ and $\gamma\Sigma$ respectively.

2. Link to Chaudhary, Fu & Li, 2023: "Corporate Bond Multipliers: Substitutes Matter"

- Both papers find higher multipliers at higher levels of aggregation.
  1. This paper: Equities

$$\Delta P_{it} = M_{idio} * (\Delta Q_{it} - \Delta Q_{group(i),t}) + M_{group} * \Delta Q_{group(i),t} + \epsilon_{i,t}$$

  2. Chaudhary, Fu & Li (2023): Corporate Bonds

$$\Delta P_{it} = M_{idio} * \Delta Q_{it} + M_{group} * \Delta P_{group(i),t} + \epsilon_{i,t}$$

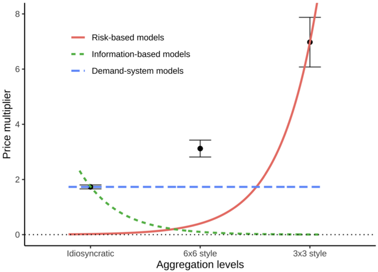
  ... using $\Delta Q_{group(i),t}$ as an instrument for $\Delta P_{group(i),t}$.
- I think the papers are complementary, but a reference (at least in a footnote) would be useful.
  - In CARA model M_{group} captures the off-diagonal elements of versus Σ^{-1} and Σ respectively.
  - Generally, 1. identifies cross-multipliers and 2. identifies scaled cross-elasticities.

3. Interpretation of the findings: Arbitrage Risk

- Proposed explanation: Compensation for bearing arbitrage risk
  - Lower for idiosyncratic mispricing. High when demand shocks loads on common factors in **covariance matrix**.
  - These factors do not necessarily have to be priced.
- Test: Multipliers should **not** be larger for factors that don't explain covariance matrix well.
  - Typically **industry factors** are not priced but explain covariance matrix better (BARRA Model)
  - Seems like an ideal candidate to test the proposed risk-based explanation.
- Alternative explanation: Mandates/Constraints.
  - Easy to drop individual small cap stock from portfolio $\rightarrow M_{smallcap}$ low
  - Harder to drop AAPL from portfolio $\rightarrow M_{AAPL}$ larger
  - Very hard to switch from value to growth $\rightarrow M_{style}$ even higher
  - Extremely hard to switch out of equities $\rightarrow M_{Mkt}$ very high

4. Calibration of CARA model: Volatility of CFs vs Returns

- "Risk-based models predict vanishingly small price multipliers at idiosyncratic levels"



- Simple CARA model with factor structure in payoffs $\frac{M_{style}}{M_{idio}} = \frac{N_s \sigma^2_{style}}{\sigma^2_{idio}}$
- $\frac{N_s \sigma^2_{style}}{\sigma^2_{idio}}$ is **much** higher than the ratio of coefficients $\frac{6.9}{1.7} \approx 4$
- You calibrate σ^2 with return volatilities as opposed to cashflow volatilities. Because $N_s > 200$ the ratio explodes?

Conclusion

- Very nice paper!
  - Contributes to our understanding of **multipliers**, which are an important new moment for testing asset pricing models.
  - Intuitive and **very robust** finding: Multipliers are larger at more aggregate levels.
- Small suggestions
  - Semantics: Informed/uninformed $\rightarrow$ Common/Investor-specific Shifts
  - Connection to Chaudhary, Fu & Li (2023)